Microwave Transmission Approach for Human Neuronal Activities Detection

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***Abstract*-Here we present a new detection approach based on microwave technology. The previous works show that the proposed approach gives a hope for human neuronal activities detection and has its own unique advantages and characteristics of non-contact and being nondestructive. In this paper, The novel detection method and its basic principle will be studied and demonstrated first. Then, an anatomically accurate high-fidelity human head model was applied in simulation to study the electric field distribution and brain's influence on the antenna radiation.**

I. INTRODUCTION

Over the past decade, the exploitation of microwave in bio-medical applications has received growing attention, because of their non-ionization and non-contact, especially for microwave imaging and detection. A microwave system has been utilized for head imaging to detect brain injuries [1],which succeeded to get the desired 2-D image of the head phantom. In the aspect of breast cancer detection and stroke diagnosis, it's also available for microwave technology to acquire the tumor location and differentiate hemorrhagic from ischemic stroke in acute stroke patients [2-4]. Due to the fact that antenna's performances have a great effect on the correction and accuracy of measurement, antennas with various performances and placement are implemented in microwave test system [5-7]. Also, it has been recently researched that microwave can get a hope to monitor dielectric properties of cerebral spinal fluid, making it possible to detect Alzheimer's disease up to a decade before the onset of symptoms [8].

The microwave detection technique in most of studies is dependent on dielectric contrast between objects of interest and its surroundings. Thus the early warning of diseases mainly relies on the structural aberration of biological tissue. Although microwave plays an increasingly important role in medical services, its application in neuronal activities detection has only recently been discovered [9].

In this paper, the microwave transmission method we proposed for detection of neuronal activities attaches importance to phase variation of electromagnetic(EM) wave propagating through brain functional site. The technique aims to extract neuronal activities from phase variation of transmission coefficient. This idea has received support from our pilot study, in which the experiment of brain neuronal activities detection on the rat reveals that phase variation of EM wave has good consistency with electroencephalography (EEG) signal [9]. Microwave transmission method proposed for neuronal activities detection has its advantages on non-contact and low-cost properties in contrast to scalp EEG, a traditional technique used for neuronal activities detection by recording of electrical activity along the scalp. Meanwhile, the method offers possibilities for non-invasive detection of deep brain activities, which can only be read via implanted EEG. The motivation for this project is from the challenge to trace the work of psychotropic drugs and monitor the brain activities of driver by a compact and wireless system, such as drowsy driving and drunk driving.

II. PRINCIPLE ANALYSIS

The detection is based on measurements and analysis of EM signals transmitted through the brain. According to existing research, the ion concentration of the extra-cellular fluid that surrounded by the neurons of the brain varies with neuronal activation [10]. This process results in dynamic nature of electric properties(including permittivity and conductivity) at brain functional site. As shown in Fig. 1, once microwave transmitted by implemented antenna travels through the brain functional site of interest, its characteristics of amplitude and phase will be changed with variation of electric properties of activated area. Thus, the activity of functional site can be known by transmission coefficient -S_{21}.

Three-layered simplified infinite slab model is utilized for propagation analysis, as depicted in Fig. 2. According to theory of EM fields, electric field at the medium ① can be written as the form:

$$\vec{E}_1(z) = \hat{e}_x E_{1im} e^{-\gamma_1 z} . \tag{1}$$

then the electric field at the interface D is

$$\vec{E}_3 = \hat{e}_x E_{1im} \tau_1 \tau_2 e^{-\gamma_3 d_l} e^{-\gamma_2 d_2} e^{-\gamma_1 d_l} . \tag{2}$$

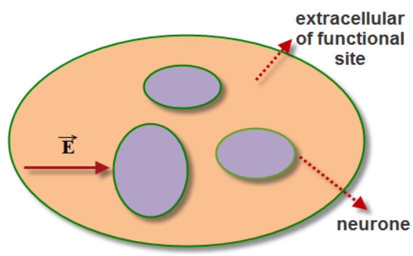

Fig 1. Microwave traveling through brain functional site.

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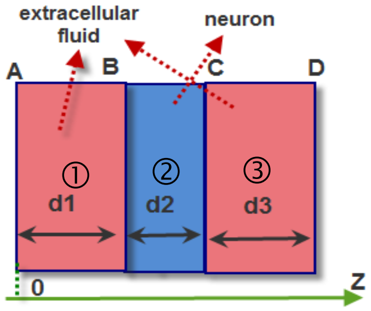

Fig 2. Simplified planar model of a single neuron for functional site.

Where γ_i, given by $j\omega\sqrt{\mu\varepsilon_{ic}}$, is the propagation constant of the ith layer, ε_{ic} is called complex dielectric constant of the ith layer, and valued by $\varepsilon_i - j\sigma_i/\omega$, and τ_1, τ_2 are the propagation coefficient of interface B and C respectively.

When the microwave travels through a mass of neurons, the phase displacement $\Delta\varphi$ between transmitting and receiving antennas-transmission coefficient, will be

$$\Delta\varphi = \sum_{i=1}^{n} \gamma_i(\varepsilon,\sigma)d_i + \sum_{j=1}^{m} \mathrm{Arg}\left[\tau_j(\varepsilon,\sigma)\right]. \quad (3)$$

Where d_i is the propagating path length of the ith layer, and $\mathrm{Arg}(\tau_j)$ is the phase of the propagation coefficient of the interface. It's important to note that both γ_i and τ_j are functions of the permittivity and conductivity.

Obviously, phase displacement is proportional to three factors: frequency of EM wave, variation of electrical characteristics and corresponding propagation path. Despite of the fact that total ion numbers of the functional site are stable during the process of neuronal activities, the phase of transmission coefficient still varies with the neuronal activities.

III. Simulation Analysis

The EM interaction between antenna and head should be firstly taken into consideration for the improvement of detection. Then, an anatomically accurate head model and EM simulation software-FEKO has been utilized to help get insight into this interaction analysis. The head model can be found in [11] and dielectric properties of tissues were obtained from [12]. Simulation diagram was illustrated in Fig. 3.

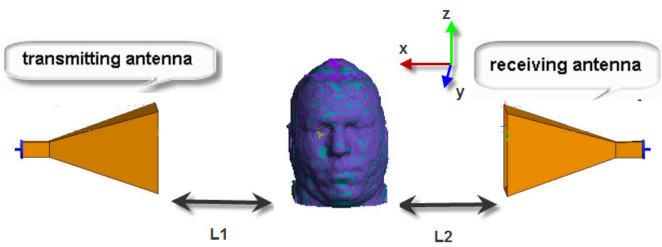

Fig 3. Simulation setup.

Standard horn antenna with a gain of 20 dB from 3.95 GHz to 5.85 GHz, are adopted for our simulation. In order to reduce the impact on radiation, the head model is suggested to be located in radiating near-field region of transmitting antenna. The aperture size of horn antenna is 21cm×27cm. Range of radiating near-field region in the 4 GHz is 32cm<R<194.4cm. Fig.4 below presents the distribution of electric field in brain, when transmitting antenna is nearly 32 cm away from head.

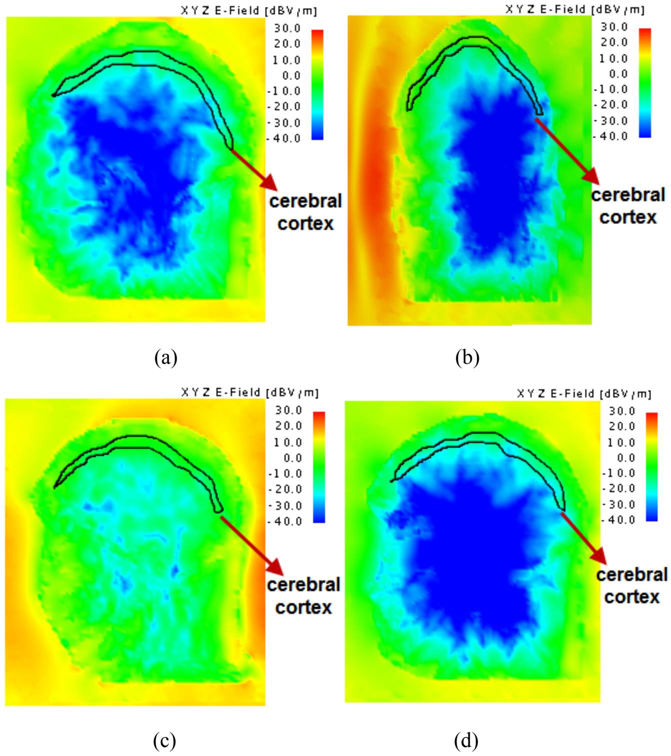

Fig 4. Electric field distribution within the brain.(a) brain center in sagittal plane.(b) brain center in coronal plane.(c) 3cm distance on the left side from brain center in sagittal plane(middle of left eye).(d) 3cm distance on the right side from brain center in sagittal plane(middle of left eye)

It can be noticed that an obvious attenuation appears when EM wave propagates through the surface of brain. The abrupt decline of electric field intensity is caused by the large contrast of dielectric constant of air, skin and subcutaneous fat. The relative permittivity is 1, 36.59, 5.13,respectively.

Nevertheless, appreciable intensity of electric field can still be found in cerebral cortex, which is represented by the area formed with black line. The result demonstrates potential capability of microwave to detect neuronal activities in the cerebral cortex. On the other hand, microwave is still capable of penetrating through our deep brain in spite of reflection and attenuation, making it possible to "read our deep thoughts".

Due to the fact that existence of head will inevitably have an influence on radiation of antenna. Radiation pattern with the different distances between head and antenna has been researched and top view of radiation pattern at 2cm distance is depicted in Fig.5. Other results are summarized in Table I.

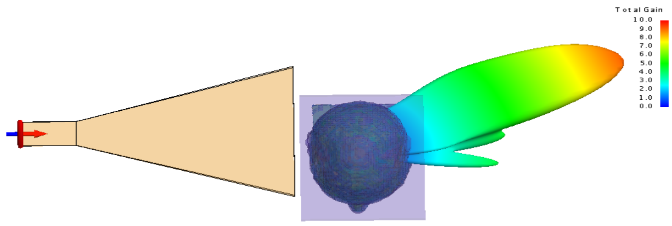

Fig 5. Radiation pattern of antenna with the distance of 2 cm.

TABLE I

RADIATION CHARACTERISTIC OF ANTENNA WITH DIFFERENT DISTANCE OF HEAD

Distance (cm)	Beam direction (Phi/degree)	reflection coefficient(dB)	Gain (dBi)
2	26	-11.08	9.50
5	-5	-16.98	7.81
	30		8.91
	82.1		6.84
8	25	-16.84	10.3
32	-2	-17.76	13.5
	21		12.84
100	12	-22.94	16.09
	0		17.56
Without head	0	-25.45	19.14

As shown in Fig. 5 and Table I, when the head is close to radiator, radiation characteristics will be affected to different extent. At the range of 1-32cm, main lobe direction is at about 25 degree. If the head is located in radiating near-field region, main lobe direction will gradually move to 0 degree as the distance increases. Meanwhile, multiple beams will arise at some particular distance. Therefore, beam selection and location of receiver are worth of serious consideration for the specific brain functional site detection such as primary visual cortex.

IV. CONCLUSION

A new method for human neuronal activities detection has been presented in this paper. It was shown that brain functional site is a kind of time-varying media, making it possible to detect neuronal activities by microwave. In addition, an accurate head model is introduced for our simulation to study distribution of electric field and brain's influence on antenna radiation. The previous works encouraged us to implement the clinical experiment for human brain activity detection. Clinical trials have recently commenced and results will be presented in future publications.

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